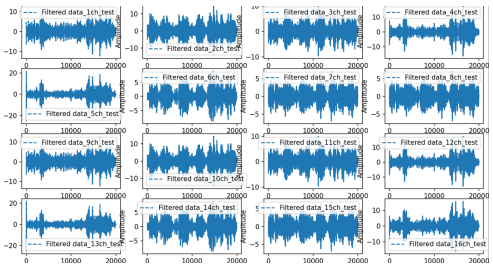
DATA INTERFACE & SOFTWARE GUIDE · SKU CQA240229G16

Dual-Chip SPI, Registers & SDKs

How PiEEG-16 talks to the Raspberry Pi: two ADS1299 chips on one SPI bus, two 27-byte frames per sample, conversion to microvolts, default registers, PiEEG Server, the Python SDK and the open datasets.



SPI0	2 × 27 B	250/s	0.268	LSL
Shared bus · mode 1	Frames per sample	Samples per second	µV per LSB (gain 1)	To MNE, OpenViBE, BCI2000

01 SOFTWARE OPTIONS

Option	Best for	Language	Start
PiEEG Server	Dashboard, recording, WebSocket, LSL, OSC, webhooks	Any (WebSocket JSON)	`pieeg-server`
Python SDK	Learning how two ADS1299 chips work together, raw data	Python	`pieeg16_drdy_only.py`
LSL	OpenViBE, MNE, BCI2000 and other LSL tools	Any	`pieeg-server --lsl`

NOTE BrainFlow.
BrainFlow's `PIEEG_BOARD` supports the 8-channel PiEEG only. Stream PiEEG-16 into other tools over LSL from PiEEG Server.

02 SPI INTERFACE

Devices	`/dev/spidev0.0` chip 1 · `/dev/spidev0.1` chip 2
Mode	1 (CPOL 0, CPHA 1)
Bit order	MSB first, 8-bit words
Clock	4 MHz (SDK); Server 4 MHz Pi 4, 2 MHz Pi 5; 1 MHz if glitches
Chip select	CE0 for chip 1 · GPIO 19 (software) for chip 2

DRDY 1	GPIO 26 · chip 1
DRDY 2	GPIO 13 · chip 2
Pi 4 chip	`/dev/gpiochip0`
Pi 5 chip	pinctrl-rp1 (gpiodetect)

03 DATA FRAMES



Figure 1 — Each chip delivers a 27-byte frame per sample. Chip 1 gives channels 1–8, chip 2 gives 9–16. The SDK accepts a sample only when chip 2's status word is 0xC0 0x00 0x08.

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$\mu V = raw \times 4.5 / 16\,777\,215 \times 10^6$

One count ≈ **0.268 µV** at gain 1. Same conversion for both chips.

READING ONE SAMPLE

```
wait_for_drdy(drdy1); frame1 = spi1.readbytes(27)   # ch 1-8
wait_for_drdy(drdy2); cs2.set_value(0)
frame2 = spi2.readbytes(27); cs2.set_value(1)       # ch 9-16
```

04 ADS1299 COMMANDS AND DEFAULT REGISTERS

Command	Code
WAKEUP	0x02
RESET	0x06
START	0x08
STOP	0x0A
RDATAC	0x10
SDATAC	0x11

Addr	Register	Value	Meaning
0x01	CONFIG1	0x96	250 SPS
0x02	CONFIG2	0xD4	Test-signal settings
0x03	CONFIG3	0xFF	Internal reference and bias buffers on
0x04	LOFF	0x00	Lead-off off
0x05–0x0C	CH1SET–CH8SET	0x00	Normal input, gain 1
0x14	GPIO	0x80	
0x15	MISC1	0x20	SRB1 on: shared reference
0x17	CONFIG4	0x00	Continuous conversion

Both chips get the same settings, chip 1 first. Write `[0x40 | addr, 0x00, value]`, read `[0x20 | addr, 0x00, 0x00]`; for chip 2 pull GPIO 19 low around each transfer. Start-up: WAKEUP, STOP, RESET, SDATAC, registers, RDATAC, START.

05 PIEEG SERVER

SHELL

```
pieeg-server                              # stream + dashboard (same as --device pieeg16)
pieeg-server --filter                     # 1-40 Hz band-pass
pieeg-server --record session.csv         # record while streaming
pieeg-server --lsl                        # Lab Streaming Layer outlet
pieeg-server --osc                        # VRChat OSC bridge
pieeg-server record session.csv --duration 300
pieeg-server monitor                      # live view in the terminal
pieeg-server doctor                       # check SPI, GPIO and settings
```

Address	What it is
`http://raspberrypi.local:1617`	Browser dashboard
`ws://raspberrypi.local:1616`	WebSocket stream: JSON frames with 16 channels in µV

06 PYTHON SDK

Script	What it does
`GUI/pieeg16_drdy_only.py`	**Start here.** Live 16-channel graph; reads each chip on its own DRDY line
`Save_data/Save_data_OS.py`	Save raw µV for all 16 channels to Excel
`Save_data/1.Save_Data.py`	Save with a live graph
`Save_data/command_line/pieeg_to_csv.py`	Command-line CSV recorder (--fname, --duration, --fs, --gain)
`Dataset/Remove_Spikes.py`	Remove SPI glitch spikes, keeping all 16 channels aligned

INSTALL AND RUN

```
sudo apt install -y python3-venv python3-dev
python3 -m venv --system-site-packages ~/pieeg-env && source ~/pieeg-env/bin/activate
pip install spidev matplotlib scipy pandas openpyxl gpiod==1.5.4
git clone https://github.com/pieeg-club/PiEEG-16.git && cd PiEEG-16/GUI
python pieeg16_drdy_only.py      # set gpiod.chip(...) to your Pi's GPIO chip
```


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07 OPEN DATASETS

File	Content	Length
1.Chewing_Blinking.xlsx	Chewing 4-3-2-1 times, then blinking 4-3-2-1 times	10,000 samples (40 s)
2.Chewing_Blinking.xlsx	A second recording of the same test	10,000 samples (40 s)
3.Alpha_test.xlsx	Eyes closed 5 s / open 5 s, repeated	20,000 samples (80 s)

Raw µV, 16 channels, 250 SPS, dry Ag/AgCl electrodes without gel. ML example (alpha-power classification) in ML_Application.

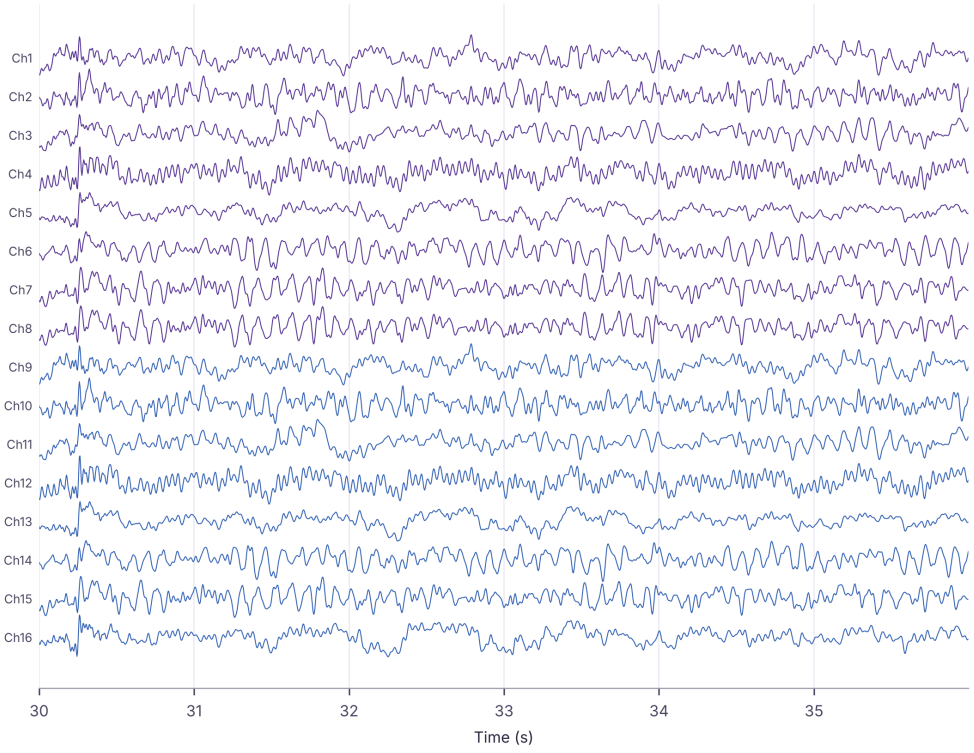


Figure 2 — 6-second excerpt of 3.Alpha_test.xlsx, glitch-cleaned, 50 Hz notch + 1–40 Hz band-pass; each trace auto-scaled.

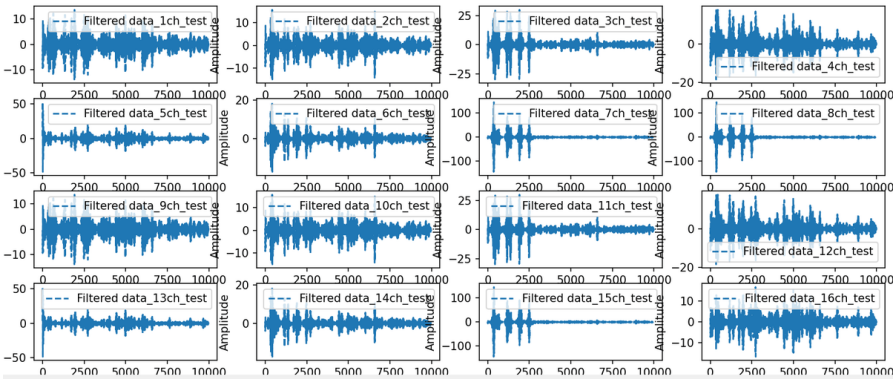


Figure 3 — Chewing and blinking artifacts on 16 channels (2.Chewing_Blinking.xlsx, 1–40 Hz).

08 RELATED DOCUMENTS

PE16-DS-001 Product Datasheet	**PE16-MD-002** Mechanical & Dimensional Drawing	**PE16-PCB-003** PCB & Manufacturing Specification
PE16-PIN-004 Connector Pinout & Electrode Guide	**PE16-QSG-005** Quick Start & Safety Guide	**PE16-ENC-007** 3D-Printed Enclosure Guide